Diagrama de clases — Procesamiento Downlink y ADR

frame

DownlinkFrame
-mhdr : String = "60"
-devAddr : String
-fCnt : String
-fPort : String
-frmPayload : String
-mic : String = "0000"
-ack : boolean
–
+toPayloadString() : String
+isAck() : boolean

crea

network

«resumen»
NetworkServer
-adrManager : AdrManager
-downlinkProcessor : DownlinkProcessor
-adrEnabled : boolean
–
+receiveFromGateway(...) : void

DownlinkProcessor
-classAPolicy : DownlinkPolicy
-classBPolicy : DownlinkPolicy
-classCPolicy : DownlinkPolicy
-ackBuilder : AckDownlinkBuilder
–
–
+process(device : Device, session : DeviceSession, gateway : Gateway, context : UplinkContext, transport : String) : void
-buildAdrCommand(session : DeviceSession) : String
-buildDownlinkCommand(ackRequired : boolean, adrCommand : String, decision : DownlinkDecision) : String
-resolvePolicy(device : Device) : DownlinkPolicy

actualiza estado ADR

«resumen»
UplinkContext
-fields : Map<String, String>
-deviceId : String
-confirmed : boolean

consulta uplink

consulta ADR / ACK

declarado

crea

DeviceSession
-currentSf : int
-recommendedSf : int
-ackRequired : boolean
-fCntDown : int
-confirmedMessages : int
-ackGenerated : int
-ackReceived : int
–
+nextFCntDown() : int
+isAckRequired() : boolean
+setAckRequired(ackRequired : boolean) : void
+getCurrentSf() : int
+getRecommendedSf() : int
+setRecommendedSf(recommendedSf : int) : void
+registerAckGenerated(fcnt : int) : void

incrementa FCntDown

AckDownlinkBuilder
+buildAck(device : Device, session : DeviceSession) : DownlinkFrame

AdrManager
+MIN_SF : int = 7
+MAX_SF : int = 12
+LOW_MARGIN_THRESHOLD_DB : double = 3.0
+HIGH_MARGIN_THRESHOLD_DB : double = 6.0
–
+recommendSpreadingFactor(linkMarginDb : double, currentSf : int) : int

obtiene DevAddr

Componente declarado en DownlinkProcessor.
El flujo actual construye DownlinkFrame directamente dentro de process().

network.policy

ClassADownlinkPolicy
+evaluate(device : Device, fields : Map<String,String>, transport : String) : DownlinkDecision

ClassBDownlinkPolicy
+evaluate(device : Device, fields : Map<String,String>, transport : String) : DownlinkDecision

ClassCDownlinkPolicy
+evaluate(device : Device, fields : Map<String,String>, transport : String) : DownlinkDecision

DownlinkPolicy
+evaluate(device : Device, fields : Map<String,String>, transport : String) : DownlinkDecision

retorna
retorna
retorna

DownlinkDecision
-shouldSend : boolean
-command : String
-fPort : String
-adrCommand : String
–
+none() : DownlinkDecision
+send(command : String, fPort : String) : DownlinkDecision
+adr(adrCommand : String) : DownlinkDecision
+shouldSend() : boolean
+getCommand() : String
+getFPort() : String
+getAdrCommand() : String

usa decisión

consulta

sendDownlink

selecciona política

entidades

«resumen»
Device
-config : LoRaConfig
-confirmed : boolean
-currentSpreadingFactor : int
–
+setSpreadingFactor(spreadingFactor : int) : void

«resumen»
Gateway
+sendDownlink(payload : String, transport : String) : void

DeviceClass
CLASS_A
CLASS_B
CLASS_C

Símbolo	Significado
–>	Asociación / referencia estructural
..>	Dependencia de uso
*–	Composición
Realización	Implementación de interfaz
+	Público
-	Privado